

Attaboy ATi

We got our hands on the hottest new graphic scorchers from ATi, and here is how they fared

ATI has launched their X800 series of video cards, putting to rest the long awaited anticipation of their sequel to the Radeon 9700 series. New technologies such as GDDR3, co-developed by ATi and Joint Electron Device Engineering Council (JEDEC), along with massive changes and improvements in the R300 graphics core, have already been implemented in this new series, bringing real-time cinematic effects closer to consumer desktops.

ATI sent us engineering samples of the X800 XT Platinum and the X800 PRO. These engineering samples are pre-mass production cards, sent to various reviewers to test and give users an insight into the their architecture. Remember, there may be differences in the performance of engineering samples and the end product you'll actually buy—not necessarily worse, the end product could actually perform better!

The X800 XT features 16 pixel pipelines, while the PRO version has 12 pipelines. The R420 core contains approximately about 160 million

transistors, and uses the 0.13 micron manufacturing process.

The power consumption of this card is on par with the older 9800 XT and PRO models. The core clock speed of the XT and the PRO models are 525 MHz and 475 MHz respectively.

Both use 256 MB of GDDR3, running at 560 MHz (1120 MHz DDR) for the X800 XT, and 450 MHz (900 MHz DDR) for the X800 PRO.

The number of VertexShaders in the X800 has gone up to six, from the previous four available on the 9800XT, and it supports VertexShader 2.0. The memory interface is divided into four channels of 64 bits (total 256 bits) that read and write data to and from the memory at the same time—controlled by a sequencer logic circuit for optimum throughput.

nVidia's offering, the 6800 or NV40, introduces shader model 3.0 features (DX 9.0c)—missing in the ATi X800 series. ATi claims that since current generation games are yet to implement PlayStation 2.0 features, it will take over a year for PlayStation 3.0 to be implemented. This seems feasible, but we feel nVidia is a better future-proof solution.

Also, the 6800 is a gigantic chip, with about 222 million transistors, and requires a hefty SMPS to power it. The original NV40 runs at a core clock speed of 400 MHz, with approximately the same memory clock speed (1.1 GHz) as the R420.

We tested the X800's on both the latest games, as well as older ones. Our test-bed was an Intel D875PBZ motherboard with a 3.2 GHz Pentium 4, 512 MB of DDR 400 RAM and a 120 GB Seagate SATA hard drive. We used the latest Forceware 56.72 drivers for the 5950 Ultra.

We started with an amazing game called *Far Cry*. Putting together a PC that can shore up to its system requirements costs as much as buying

a car. We ran the default fort time demo at both, maximum and minimum details. This game is one of the first to implement DirectX 9.0 features such as PS2.0. Even the X800 couldn't keep up when everything was turned on.

We also tested *Halo* and *Unreal Tournament 2004* with



Beastly beauties: the X800 XT and PRO cards

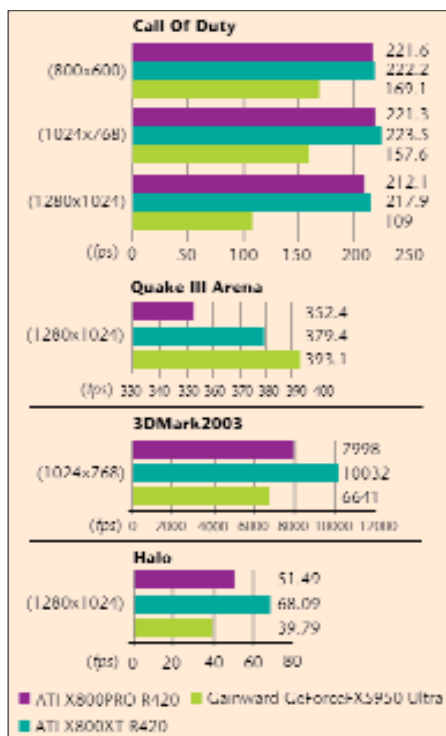


the default benchmarks. One major problem is that the CPU seem to become a bottleneck—evident from the fact that when we changed the resolution from 800 x 600 to 1024 x 768, in *Call Of Duty*, the difference in frame-rates is negligible, but *Halo* and *Unreal Tournament 2004* show a drop in frame rates.

In the *Halo* tests, the X800XT stood out with a stellar performance, as compared to the GeForce FX 5950Ultra, which struggled at lower resolutions. *Call of Duty* uses a heavily modified *Quake III* engine. This game gave frame rates upwards of 200 with both the X800 cards, while it languished at 109 fps with the 5950—the X800XT out performed it by 88 per cent at the highest tested resolutions.

ATI has a winner on its hands with the X800 series. The X800 XT and the X800 PRO are two of the fastest cards we've ever tested. Both cards performed similarly, but you'll need a high-end machine to enjoy these cards to the fullest, since the CPU and the system RAM can easily become bottlenecks.

Other mainstream models are in the offing, and the pricing of the cards will vary as mainstream and performance cards will have different specifications. ATi has once again set the bar with their X800 series. For now, we're dying to see the nVidia 6800 do battle with ATi's X800! 



*These games were run at highest quality settings with 32-bit colour

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